

Chapter 1

Mesh generation based on Geompack++

In this document we demonstrate how to generate quadrilateral `oomph-lib` meshes, based on the output from Barry Joe's mesh generator `Geompack++`, available as freeware at <http://members.shaw.ca/bjoe/>. The mesh generation is performed in a two-stage process. First we use `Geompack++` to generate the mesh "offline". Then we process the output files generated by `Geompack++` to generate an `oomph-lib` mesh.

1.1 Example of the use of Geompack++

The documents `meshoper.pdf` and `regmesh.pdf` distributed with `Geompack++` contain a comprehensive User's Guide for the code and its many options. `Geompack++` is a very sophisticated mesh generator and can be used to create 2D and 3D meshes with a wide variety of element types. Here we only discuss how to use `Geompack++` to generate 2D quadrilateral meshes. [The development of `oomph-lib` meshes that process `Geompack++`'s output for other element types should be straightforward but is still on our "To Do" list – any volunteers?]

`Geompack++` creates quadrilateral meshes, based on the information about the mesh boundaries (nodes and curves) provided in two input files, `filename.rg2` and `filename.cs2`, say. An output file, `filename.1.mh2` is created. It contains the information about the mesh boundaries, the nodal positions and the element connectivity lists.

1.1.1 How to visualise a mesh generated by Geompack++

To visualise the mesh, the program `Geomview` is available. (`Geomview` can be downloaded from the [Geomview home page](#).) The `Geompack++` output file cannot be processed directly by `Geomview` but a simple program that converts the file `filename.mh2` to a format that is readable by `Geomview` is [available](#).

1.1.2 The example: A rectangle with a hole

As an example we demonstrate how to use `Geompack++` to generate a mesh for the rectangular domain with a hole shown in the figure below. The domain is defined by eight nodes and eight curves (straight lines) which connect the nodes and define two boundaries.

The boundary numbers shown in the sketch correspond to those in the `Geompack++` input region file `box_hole.rg2` and in the curve input file `box_hole.cs2`. In the corresponding `oomph-lib` mesh, the boundaries are numbered from zero.

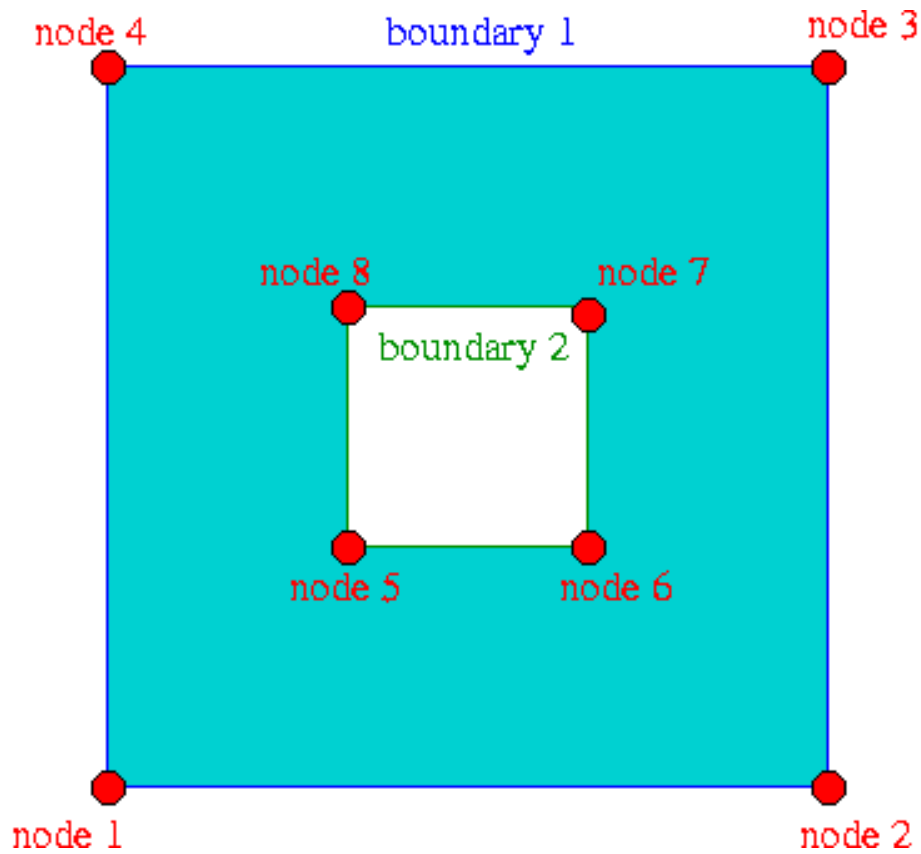


Figure 1.1 A rectangular domain with a hole.

The desired mesh characteristics are defined in the file `box_hole.m2` which specifies that the final mesh should contain approximately 200 elements. When processed with the command

```
./geompack++ box_hole.m2 error.log
```

`Geompack++` creates the output file `box_hole.mh2`.

Here is a sketch of the resulting discretisation, as displayed by `geomview`:

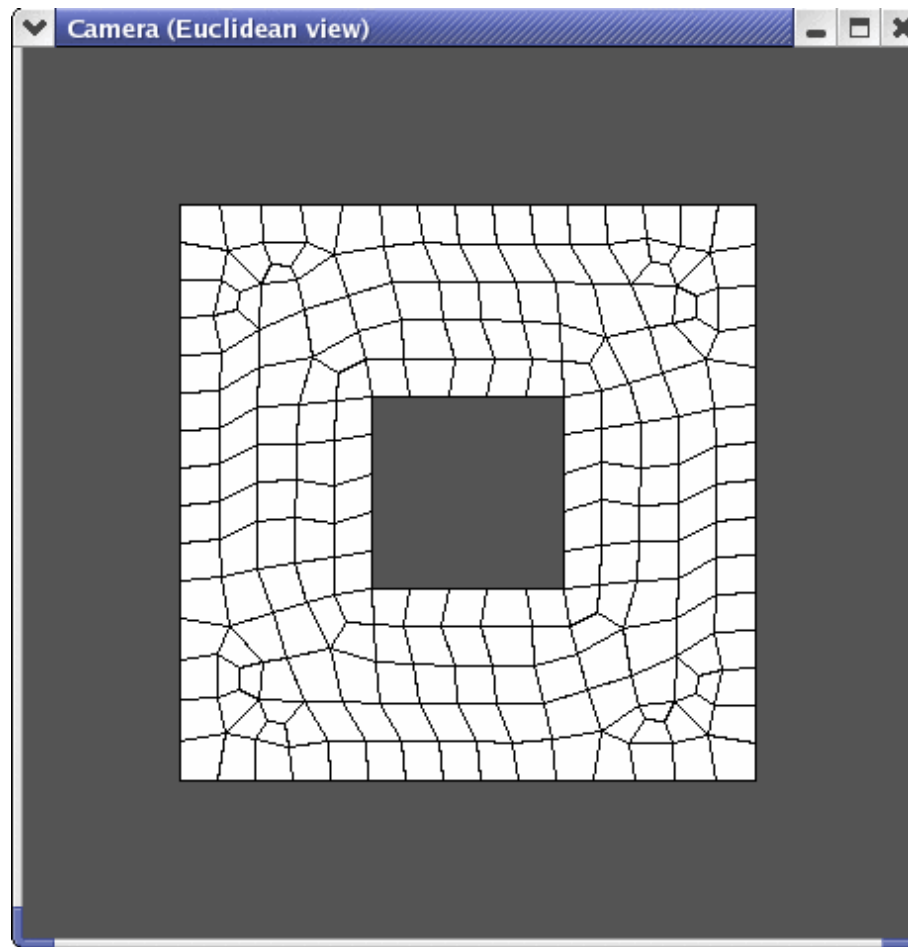


Figure 1.2 Screenshot of geomview, showing the discretisation of the rectangular domain with a hole.

1.2 Creating an oomph-lib mesh based on output files generated by Geompack++

oomph-lib provides a mesh, `GeompackQuadMesh`, that uses the output from `Geompack++` to generate an oomph-lib Mesh containing `QElement<2, 2>` four-node quadrilateral elements. The relevant interface is:

```

//=====start_of_geompackquadmesh_class=====
/// Quadrilateral mesh generator; Uses input from Geompack++.
/// See: http://members.shaw.ca/bjoe/
/// Currently only for four-noded quads -- extension to higher-order
/// quads should be trivial (see the corresponding classes for
/// triangular meshes).
//=====
template<class ELEMENT>
class GeompackQuadMesh : public Mesh
{
public:
    /// Constructor with the input files
    GeompackQuadMesh(const std::string& mesh_file_name,
                    const std::string& curve_file_name,
                    TimeStepper* time_stepper_pt = &Mesh::Default_TimeStepper)

```

The driver code `mesh_from_geompack_poisson.cc` demonstrates the use of this mesh for the solution of a 2D Poisson problem in the "rectangular domain with a hole", described in the previous section.

The code expects the names of `*.mh2` and `*.cs2` files generated by `Geompack++` as command line arguments and stores them in the namespace `CommandLineArgs`

```

int main(int argc, char* argv[])
{
    // Store command line arguments
    CommandLineArgs::setup(argc, argv);

    // Check number of command line arguments: Need exactly two.
    if (argc!=3)
    {
        std::string error_message =

```

```

    "Wrong number of command line arguments.\n";
    error_message +=
    "Must specify the following file names  \n";
    error_message +=
    "filename.mh2 then filename.cs2\n";

    throw OomphLibError(error_message,
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
}

```

The names of these files are then passed to the mesh constructor. Since the rest of the `driver code` is identical to that in the `corresponding example with a structured mesh`, we do not provide a detailed code listing but simply show the plot of the computed results, together with the tanh-shaped exact solution of the problem:

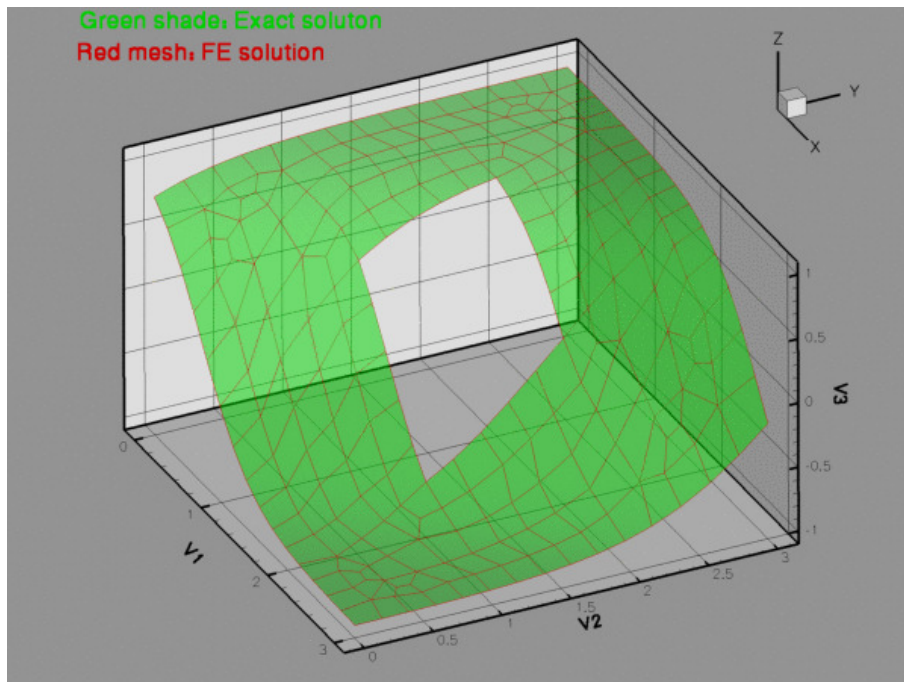


Figure 1.3 Computed and exact solutions.

1.3 Source files for this tutorial

- The source files for this tutorial are located in the directory:

`demo_drivers/meshing/mesh_from_geompack/`

- The driver code is:

`demo_drivers/meshing/mesh_from_geompack/mesh_from_geompack_poisson.cc`

1.4 PDF file

A [pdf version](#) of this document is available. \